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Insertion of hepatic lesions into clinical photon-counting-detector CT projection data

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Abstract

Objective. To facilitate task-driven image quality assessment of lesion detectability in clinical photon-counting-detector CT (PCD-CT), it is desired to have patient image data with known pathology and precise annotation. Standard patient case collection and reference standard establishment are time- and resource-intensive. To mitigate this challenge, we aimed to develop a projection-domain lesion insertion framework that efficiently creates realistic patient cases by digitally inserting real radiopathologic features into patient PCD-CT images.

Approach. This framework used an artificial-intelligence-assisted (AI) semi-automatic annotation to generate digital lesion models from real lesion images. The x-ray energy for commercial beam-hardening correction in PCD-CT system was estimated and used for calculating multi-energy forward projections of these lesion models at different energy thresholds. Lesion projections were subsequently added to patient projections from PCD-CT exams. The modified projections were reconstructed to form realistic lesion-present patient images, using the CT manufacturer's offline reconstruction software. Image quality was qualitatively and quantitatively validated in phantom scans and patient cases with liver lesions, using visual inspection, CT number accuracy, structural similarity index (SSIM), and radiomic feature analysis. Statistical tests were performed using Wilcoxon signed rank test.

Main results. No statistically significant discrepancy ($p > 0.05$) of CT numbers was observed between original and re-inserted tissue- and contrast-media-mimicking rods and hepatic lesions (mean $\pm$ standard deviation): rods 0.4 ± 2.3 HU, lesions -1.8 ± 6.4 HU. The original and inserted lesions showed similar morphological features at original and re-inserted locations: mean $\pm$ standard deviation of SSIM 0.95 ± 0.02 . Additionally, the corresponding radiomic features presented highly similar feature clusters with no statistically significant differences ($p > 0.05$).

Significance. The proposed framework can generate patient PCD-CT exams with realistic liver lesions using archived patient data and lesion images. It will facilitate systematic evaluation of PCD-CT systems and advanced reconstruction and post-processing algorithms with target pathological features.

Keywords

Photon counting detector; Computed Tomography; Hepatic lesions; Lesion insertion; Projection data; Radiomics

1. Introduction

Photon-counting-detector CT (PCD-CT) offers several technical advantages, including improved spatial resolution, increased radiation dose efficiency, and enhanced iodine signal, which have been translated into clinical practice, likely leading to improved diagnostic confidence, reduced interpretation errors, and new imaging capabilities across various clinical applications (McCollough et al., 2023; Sartoretti et al., 2023). Furthermore, the imaging capabilities of PCD-CT could be further enhanced when combined with advanced iterative and AI-based image reconstruction, processing, and analysis techniques (Sartoretti et al., 2022; Pinos et al., 2023; Baffour et al., 2023; Hertel et al., 2023). Nonetheless, precautions and thoughtful evaluations are necessary to identify the corresponding limitations and failure modes. Premature deployment, such as using off-the-shelf AI models without calibration to differences in patient population and hospital-specific practice, along with sub-optimal protocol settings like overly aggressive dose reduction, can substantially compromise clinical benefits (Gichoya et al., 2023; Mohammadinejad et al., 2021; Choi et al., 2022). To ensure the safety and reliability of these techniques in clinical settings, task-driven quality assessment and protocol optimization using accurately annotated patient cases with target diseases are essential.

Standard patient case collection and reference standard establishment are time- and resource-intensive (Obuchowski and Bullen, 2022). To tackle this challenge, several in-silico simulation tools were developed, using numerical modeling of human anatomy (e.g., non-uniform B-spline), lesion morphology, and a PCD-CT system (Segars et al., 2010; Sauer et al., 2023; Abadi et al., 2022). To generate disease-present cases (i.e., in-silico phantoms), simulated lesion models have been inserted into the simulated anatomy before virtual CT scans, using image-domain blending methods, such as Gaussian blurring, Poisson editing, and filtering with measured modulation transfer function and noise power spectrum (Solomon and Samei, 2014; Pezeshk et al., 2017). These tools have facilitated theoretical characterization and prototyping of CT systems and reconstruction / noise reduction algorithms. However, extensive clinical validation is required to assess if such in-silico phantoms could replace true patient data across various clinical tasks. Moreover, the emerging deep-learning-assisted image-domain lesion insertion (Gong et al., 2021) exhibited potential for efficient, high-quality, large-scale simulation that incorporates scanner-specific characteristics and adaptation to additional CT systems. Nonetheless, such methods necessitate well-curated training datasets and proper training to ensure generalizability, which can be resource-intensive. An alternative approach uses projection-domain lesion-insertion tools which digitally merge forward projection of true lesion images into true patient projection data before conducting CT reconstruction and downstream tasks (Chen et al., 2015; Ferrero et al., 2017). These tools can readily incorporate high clinical realism and technical characteristics of target CT systems and algorithms in simulated patient data

and have been validated in patient evaluation studies on both liver lesions and lung nodules (Chen et al., 2016; Ma et al., 2017). However, these prior projection data-based methods require scanner-specific proprietary information and tools dedicated to conventional energy-integrating-detector CT (EID-CT). Due to the differences in scanner-specific information and tools (e.g., data acquisition modes, raw data decoder, beam-hardening correction), these prior methods are not readily applicable to PCD-CT.

In this study, we aimed to develop and validate a projection-domain lesion-insertion framework for PCD-CT. This framework used AI-assisted lesion detection and contouring to efficiently create initial lesion annotations, which were then manually refined as needed. Multi-energy forward projections of lesion images were added to projection data from PCD-CT. This framework was evaluated using phantom scans and abdominal CT cases with benign and metastatic liver lesions.

2. Materials and Methods

2.1. The proposed lesion insertion framework

The previous projection-domain lesion insertion methods for EID-CT systems (Chen et al., 2015; Ferrero et al., 2017) were modified for a clinical PCD-CT system used in our institute. The proposed workflow is illustrated in Figure 1, with major steps summarized as follows. The manufacturers' proprietary decoder software was used to decipher the raw data off the PCD-CT scanner and retrieve projection data for each detector energy threshold. Digital lesion models were generated from real lesion images in archived patient PCD-CT exams. Since clinical PCD-CT system acquires multi-energy projection data by default, lesion insertion must be performed in spectral domain. Multi-energy forward projections of lesion models were calculated using Siddon's algorithm (Siddon, 1985), considering only the primary x-ray beam. This simplification was applied due to the limited lesion size. Moreover, commercial scatter correction techniques of PCD-CT system were effective across a typical range of patient habitus and clinical scan protocols. (Rajendran et al., 2022; Liu et al., 2024; Rybertt et al., 2025) Therefore, the residual scattering effects in lesions were deemed to be negligible. To create lesion-present projections, the simulated lesion projections were digitally added to real, lesion-absent patient projections acquired from PCD-CT scans. Therefore, any residual scattering effects in patient abdomen had been implicitly incorporated. Finally, the modified projections were reconstructed using the manufacturer's offline reconstruction software to generate patient images with the inserted target pathologic features. Further details are presented in the following sections.

2.2. Patient case collection and lesion model generation

The institutional review board of our institute approved this retrospective data use study. A supervising subspecialized abdominal radiologist and two clinical research fellows selected patients based on the following inclusion criteria: 1) patient aged 18 or older undergoing contrast-enhanced abdomen-pelvis CT scan on a clinical PCD-CT system (Siemens NAEOTOM Alpha); 2) CT projection data archived from the scanner; 3) the presence of benign and / or metastatic liver lesions. The cases that did not meet inclusion criteria were excluded from this study. The supervising radiologist established reference

standards for each lesion (for a maximum of 5 lesions per case) based on biopsy information and / or follow-up exams, using a previously-configured CT image viewing workstation (Fletcher et al., 2017). Both benign and metastatic liver lesions were annotated by drawing a circle or a line across the greatest lesion diameter. Major parameters of the CT scans and image reconstruction are summarized in Table 1.

Lesion models were digitally generated using a semi-automatic annotation and contouring process. A pre-trained AI-based liver lesion detection and segmentation model, *SwimUNETR* (Hu et al., 2023), was used to automatically create high-quality initial annotation and contours of the liver lesions in each patient case. Medical image analysts reviewed the outputs of the AI model and, if necessary, used the image segmentation module in 3D-slicer (Pinter *et al.*) to refine lesion annotations and contours (Figure 2), adhering to the reference standard established by the supervising radiologist. Of note, all medical image analysts had previously participated in a training session where they practiced lesion contouring under the guidance of radiologists. The refined lesion contours were used to segment lesion images and subsequently create individual lesion models for downstream tasks. Compared to manual contouring from scratch (Chen et al., 2015), this proposed process reduced the difficulty of lesion image preparation and enhanced workflow efficiency. Additionally, it avoided potential annotation errors compared to using AI alone for lesion labeling.

2.3. Estimation of the energy for beam-hardening correction

The CT numbers of the lesions, in Hounsfield Units (HU), were converted to linear attenuation coefficients before calculating forward projection of lesion images. A monoenergetic x-ray spectrum per detector energy threshold was applied to determine the corresponding water attenuation coefficients used in this conversion. The corresponding x-ray energy was equated to the effective beam hardening correction energy used in the target PCD-CT system (Appendix Figure 1). The rationale is that commercial beam hardening correction process had been already applied to patient projections decoded from PCD-CT raw data, allowing the forward projections of lesion models to be readily added to patient projections once the effective beam hardening correction energy was used in the conversion. Since commercial beam hardening correction process was proprietary and non-transparent to clinical users, a phantom-based method from (Chen et al., 2015) was modified to estimate true effective beam hardening correction energy E_{BHC}^{true} . Major steps are described below.

An abdomen-sized phantom (Gammex, Sun Nuclear, Inc.; lateral width 40 cm) containing rods (blood, adipose tissue, iodine and calcium) was scanned on the same PCD-CT system as used in the patient exams using the same scan protocol. The rod images were segmented and digitally re-inserted into different locations in the uniform phantom background. The corresponding forward projection per detector energy threshold was calculated using an initial estimation of beam hardening correction energy E_{BHC}^{guess} and the corresponding water linear attenuation coefficient $\mu_{H2O, E_{BHC}^{guess}}$. The actual value of E_{BHC}^{true} and the corresponding water attenuation coefficient $\mu_{H2O, E_{BHC}^{true}}$ were estimated through a calibration process summarized in the following equations:

$$\mu_{m, E_{BHC}}^{guess} = \left(\frac{HU_m^{scan}}{1000} + 1 \right) \cdot \mu_{H2O, E_{BHC}}^{guess} \quad (1)$$

$$HU_m^{recon} = \left(\frac{\mu_{m, E_{BHC}}^{guess}}{\mu_{H2O, E_{BHC}}^{true}} - 1 \right) \cdot 1000 \quad (2)$$

$$\mu_{H2O, E_{BHC}}^{true} = \frac{\mu_{m, E_{BHC}}^{guess}}{\frac{HU_m^{recon}}{1000} + 1} = \frac{\frac{HU_m^{scan}}{1000} + 1}{\frac{HU_m^{recon}}{1000} + 1} \cdot \mu_{H2O, E_{BHC}}^{guess} \quad (3)$$

where $\mu_{m, E_{BHC}}^{guess}$ denotes the initial guess of linear attenuation coefficient of the m^{th} rod material, HU_m^{scan} indicates the CT numbers HU_m^{scan} measured from real phantom scans, and HU_m^{recon} is the CT numbers measured from the re-inserted rod. Of note, E_{BHC}^{guess} was derived through a numerical process. For each rod, a two-term Gaussian model was created by fitting the mappings between theoretical monoenergetic CT numbers and x-ray energy levels (range [40, 100] keV):

$$HU_{keV} = a_1 \cdot e^{\left(-\frac{E_{keV} - b_1}{c_1} \right)^2} + a_2 \cdot e^{\left(-\frac{E_{keV} - b_2}{c_2} \right)^2} \quad (4)$$

where HU_{keV} indicates the monoenergetic CT number for a given rod material, E_{keV} indicates the corresponding monoenergetic x-ray energy, and a, b, c were free parameters of the two-term Gaussian model. These models were thereafter jointly solved to derive E_{BHC}^{guess} , using constrained non-linear multivariate solver with CT numbers measured from original rod images.

2.4. Experimental evaluation methods

Time spent on each major step of lesion insertion workflow was measured. Lesion-level performance of the pre-trained Swim UNETR model was evaluated for lesion localization and segmentation, using confusion matrix, sensitivity, precision, and dice coefficient. We did not include specificity, false positive rate, and negative predictive value in lesion-level performance evaluation, because true negatives were not well defined in this context. Moreover, the accuracy of the estimated beam-hardening correction energy was validated by comparing the mean CT numbers of original rods and re-inserted rods, using rod images from low- and high-energy thresholds. To assess validity of inserted lesions, all lesion models were digitally re-inserted into different locations within the relatively uniform liver background of original patient cases. Lesion images at 65keV were used instead of

low- and high-energy thresholds, to mitigate noise perturbation in evaluation. Of note, the manufacturer's offline reconstruction software applied iterative reconstruction with stronger denoising for VMI images but by default used standard filtered-back-projection for images at low- and high-energy thresholds. Consequently, images at 65keV were less noisy compared to those at low- and high-energy thresholds. In addition, this setting aligned with one of our institute's routine protocols, thereby enhancing the clinical relevance of the presented results. The similarity between original lesions and the inserted lesions was qualitatively and quantitatively evaluated. Qualitative evaluation involved perceptual assessment of lesion appearance and line profiles. In quantitative evaluation, structure-similarity-index (SSIM) and mean CT number difference between original lesion and the inserted lesions were assessed. Radiomic features from original and re-inserted lesions were compared using Wilcoxon signed-rank test, with a p-value < 0.05 indicating statistical significance. Radiomic features were extracted using the open-source package, *Pyradiomics* v3.1.0 (van Griethuysen et al., 2017). The corresponding feature classes included gray level co-occurrence matrix (24 features), gray level run length matrix (16 features), gray level size zon matrix (16 features), neighbouring gray tone difference matrix (5 features), and gray level dependence matrix (14 features). Quantitative analyses were performed for all lesions, as well as for separately benign and metastatic subgroups.

3. Results

3.1. Patient cohort and characteristics

We included 10 cases (6 female and 4 male). Scan dates ranged from September 2023 to January 2024. From these cases, 20 lesions (10 benign and 10 metastatic) were annotated. Patient age, body mass index, lesion diameter, and lesion contrast are summarized in Table 2.

3.2. Breakdown of time efficiency

The breakdown of time spent per major step is detailed as follows. AI-assisted lesion segmentation required roughly 5 to 10 minutes (min) per patient case during inference. When necessary, manual correction was performed, typically needed 2 to 5 min per lesion. The beam-hardening calibration process, which included rod insertion and phantom image reconstruction, took approximately 120 min. Notably, this step was performed only once prior to lesion insertion. On average, the generation of lesion projections and the overwriting of patient projections required roughly 11 min and 5 min per lesion, respectively. The offline image reconstruction averaged approximately 45 min per patient case. Of note, GPU acceleration was used only for lesion segmentation and image reconstruction, whereas other steps were executed on CPU. Computational efficiency is expected to improve with further GPU acceleration.

3.3. Performance of pre-trained lesion segmentation model

Using initial lesion contours from Swin UNETR may introduce unrealistic structures (e.g., artificial iodine enhancement or vessels-mimicking structure) at new locations (Appendix Figure 2), due to excessive liver background being included in initial contours and poor blending with new anatomical context. Moreover, the pre-trained Swin UNETR yielded

sub-optimal lesion-level performance (Appendix Table I): sensitivity 0.85, precision 0.55, and dice coefficients (mean $\pm$ standard deviation) 0.71 ± 0.28 . These results highlighted the need for additional review and manual correction to complement AI-assisted lesion segmentation.

3.4. Validation of the estimated beam-hardening correction energy

A comparison of CT numbers of original and re-inserted rods is presented in Table 3. No statistically significant difference was observed in CT numbers ($p > 0.05$): the range of CT number differences were $[-2.7, 2.8]$ HU (corresponding to $[-0.2\%, 3.8\%]$) and $[-3.4, 4.4]$ HU (corresponding to $[-1.7\%, 5.4\%]$) in images from the low and high energy thresholds, respectively. This validated the estimation accuracy of beam-hardening correction energy.

3.5. Validation of lesion similarity

In visual inspection, the inserted lesions yielded similar appearance compared to original lesions (Figure 3). The corresponding line profiles indicated comparable lesion contrast and edge sharpness of lesions at original and re-inserted locations (Figure 4). Of note, the inserted and original lesion would not appear identical, because of partial volume effects and background difference at different lesion locations. In quantitative validation, the evaluation of SSIM and CT number difference further confirmed the perceptual similarity (Table 4). No statistically significant difference was observed between the radiomic features from original and re-inserted lesions ($p > 0.05$, Figure 5a). In cluster analysis, feature clusters exhibited high similarity in radiomic feature correlation matrices from original and re-inserted lesions (Figure 5b).

4. Discussion

In this study, a projection-domain lesion-insertion framework was developed and validated for PCD-CT. This framework utilized an AI-assisted semi-automatic process to efficiently generate lesion models from real lesion images. Effective beam-hardening correction energy in PCD-CT system was numerically derived using a modified CT phantom-based calibration process. Multi-energy forward projections of lesion models were calculated and added to original patient projections. The modified projections were reconstructed to create new lesion-present images. Experimental validation was conducted with phantom scans and patient exams. The proposed framework can generate patient images with realistic radiopathologic features in PCD-CT.

Using appropriate beam-hardening correction energy, the forward projections yielded sufficient CT number accuracy in rods and lesions. We observed slightly larger CT number differences in lesions compared to phantom rods, attributed to partial volume effects and background difference between original and re-inserted lesion locations. However, morphological similarity between original and re-inserted lesions were validated in visual inspection and SSIM assessment. Furthermore, no statistically significant changes were observed across radiomic features. The patterns of radiomic feature clusters were retained after lesion insertion. These findings suggested that the proposed framework can provide sufficient image quality of inserted lesions in many applications, such as generating

experimental trials for virtual imaging trials and creating training / validation data for AI-based lesion detection models.

In the proposed method, lesion projections were synthesized using real PCD-CT lesion images along with the imaging geometry and scan trajectory of clinical PCD-CT system. Consequently, the simulated lesion projections inherently captured key effects of PCD-CT system's data acquisition process. These lesion projections were digitally added to real patient PCD-CT projections, generating realistic lesion-present projections that retained characteristics of PCD-CT. To synthesize lesion-present images, the modified patient projections were reconstructed with the manufacturer's offline reconstruction software, functionally equivalent to the commercial on-board version. In summary, the proposed method effectively preserved key characteristics of PCD-CT data in the simulated lesion-present patient cases.

Compared to prior projection-domain methods (Chen et al., 2015; Ferrero et al., 2017), the proposed framework involved several major modifications. First, it extended lesion insertion from conventional EID-CT systems to a PCD-CT system. Forward projections were calculated for two x-ray energy thresholds used in raw CT data acquisition, with the flexibility to extend calculations for additional energy threshold settings (e.g., > 2 thresholds) if needed. Second, it utilized an AI model to provide initial lesion annotations and contours rather than relying entirely on manual lesion contouring by human experts, thereby streamlining lesion model preparation process. Third, in phantom-based calibration, initial guess E_{BHC}^{guess} was numerically derived through solving two-term Gaussian models instead of using fully empirical estimation. Of note, we observed that original phantom-based calibration (Chen et al., 2015) could fall apart when the difference between E_{BHC}^{guess} and E_{BHC}^{true} was large. Therefore, the proposed estimation of E_{BHC}^{guess} was effective for improving the estimation accuracy of E_{BHC}^{true} and $\mu_{H2O, E_{BHC}^{true}}$.

The projection-domain and deep-learning-assisted image-domain methods offer distinct advantages and can be applied based on the availability of the manufacturers' proprietary information and tools. Given slow computational efficiency, the projection-domain approach is suited for generating small-to-medium-sized datasets for a specific CT system. It supports flexible reconstruction across various image types, aiding evaluation of reconstruction settings on lesion appearance and detectability. In contrast, the deep-learning-assisted image-domain method, which does not require access to proprietary projection data or tools, is more scalable and efficient for large-scale simulation. However, it relies on carefully curated training data and robust training procedures to ensure generalizability and may require fine-tuning for newer CT systems and scan protocols. Notably, the projection-domain method, if available, can provide effective fine-tuning datasets to extend the image-domain method for newer scanners and protocols.

We acknowledge some limitations in the presented work. First, the validation of the estimated beam-hardening correction energy was conducted with only one configuration of CT scan parameters (e.g., tube voltage and radiation dose). This was because the phantom scans used the same scan protocol as the patient exams. However, the corresponding phantom-based estimation method is readily applicable to additional scan protocols. Further

validation could be conducted in future studies with other CT tasks. Second, no radiologist evaluation study was performed to systematically assess the perceptual similarity between original and re-inserted lesions. Nonetheless, we conducted systematic, objective image quality assessment. Further evaluation with additional CT exams, radiopathologic features, and radiologist evaluation studies will be included in future studies.

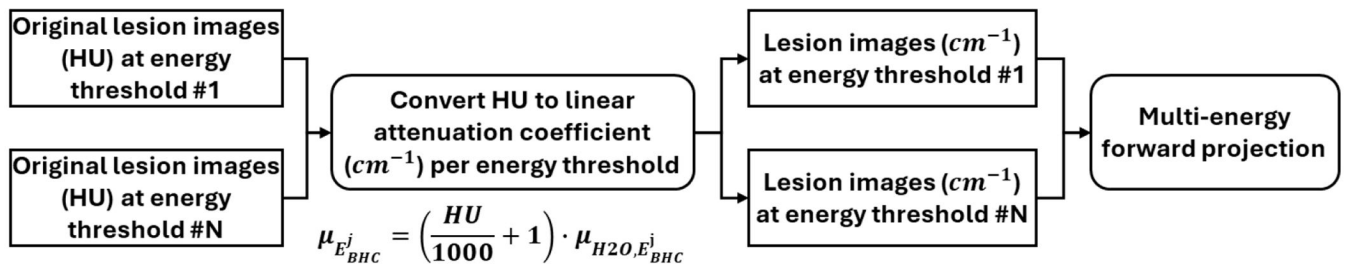
5. Conclusion

In summary, a projection-domain lesion insertion framework was developed and validated for PCD-CT system. This tool effectively reflected the effects of imaging hardware and reconstruction algorithm of PCD-CT system, allowing for generating realistic patient PCD-CT exams with target radiopathologic features.

Acknowledgements

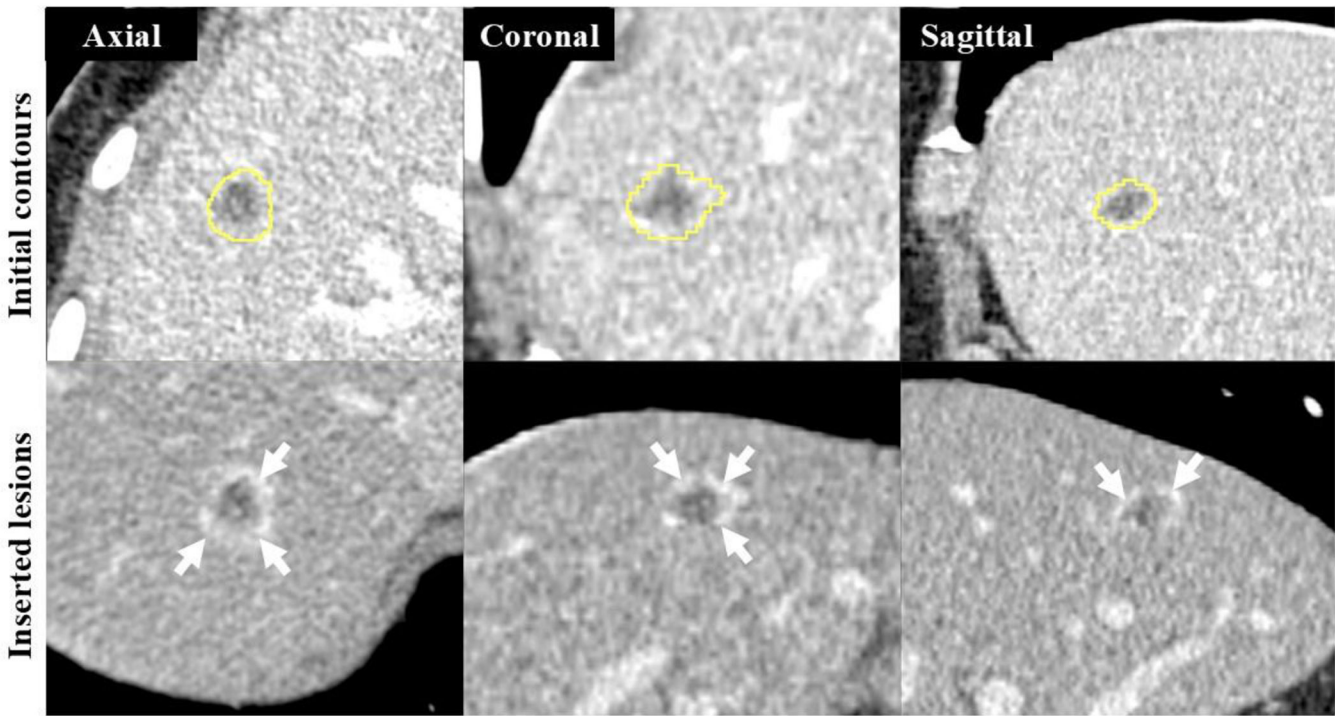
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1.: Appendix



Appendix Figure 1.

The unit conversion (from Hounsfield Unit to cm^{-1}) before conducting multi-energy forward projection across all N energy thresholds in PCD-CT. The effective beam-hardening correction energy per energy threshold (E_{BHC}^j , where $j = 1, \dots, N$) was used in the conversion.



Appendix Figure 2.
 Examples of lesion insertion using initial contours only. The initial contours included excessive liver background with higher iodine enhancement, which was carried over to the new location. Without further contour correction, this mismatch induced artifacts mimicking iodine enhancement at new lesion boundary (arrows).

Appendix Table I
 The performance of the pre-trained Swim UNETR in lesion localization *

	Predicted		Sensitivity
	Positive	Negative	
Actual	Positive	3	0.85
	Negative	14	
	Precision	0.55	

* The lesion localization performance was assessed at lesion level. In this context, true negatives were not well defined.

Data availability statement

The projection data cannot be made publicly available upon publication because they contain proprietary information. The image data that support the findings of this study are available from the authors upon reasonable request.

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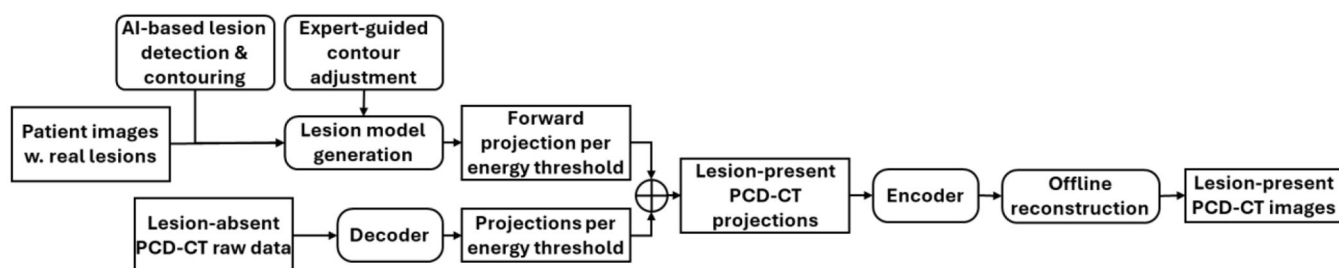


Figure 1.

The framework of the proposed projection-domain lesion insertion technique for PCD-CT.

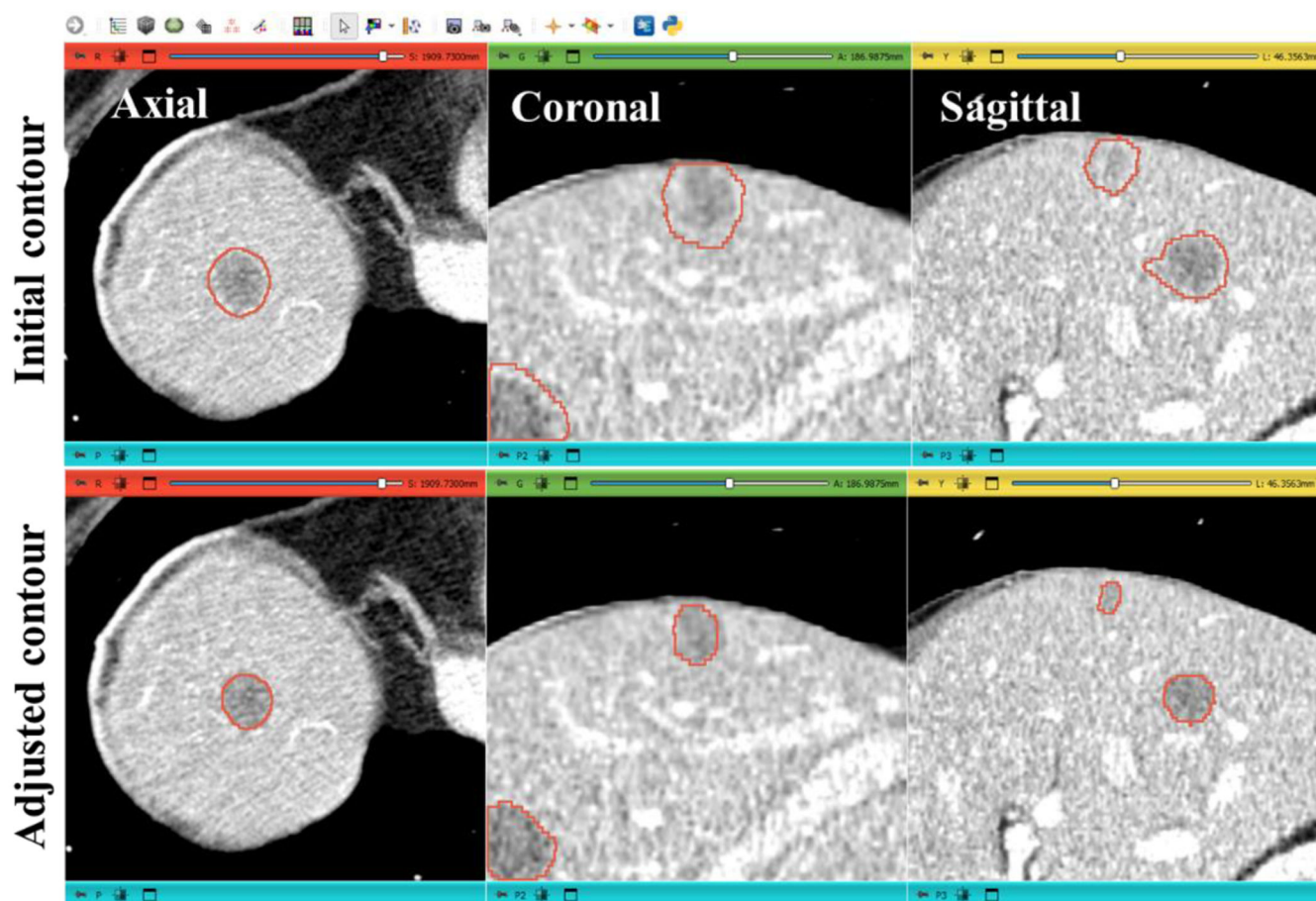


Figure 2.
Example of initial and adjusted lesion contours (i.e., red color markers) across axial, coronal and sagittal planes in 3D slicer.

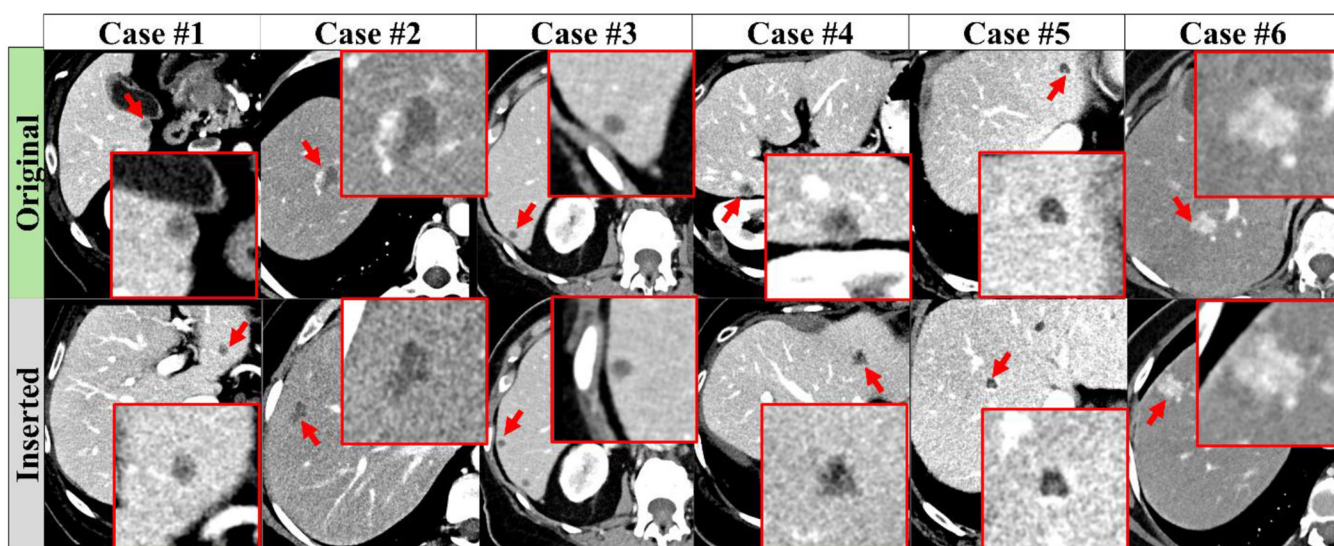


Figure 3. Examples of lesion images (65 keV VMIs) in original and re-inserted locations across different patient cases. The inserted lesions exhibited comparable appearance compared to the original ones. Arrows indicate lesion locations. The insets illustrate zoomed region-of-interests centered at the corresponding lesions.

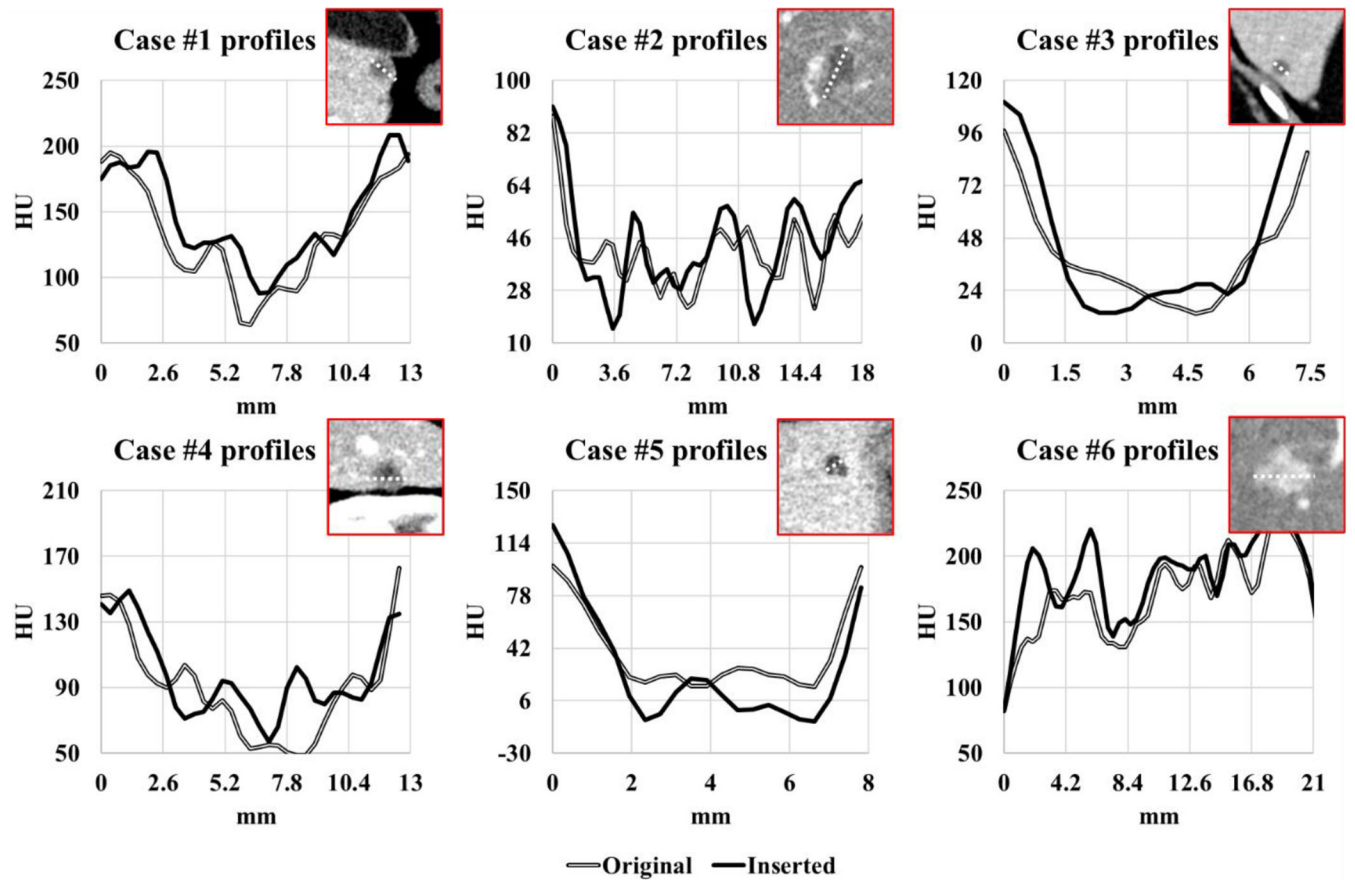


Figure 4.

Examples of line profiles of original and re-inserted lesions from Figure 3. The dotted line segments in insets indicate the locations of line profiles in original lesions.

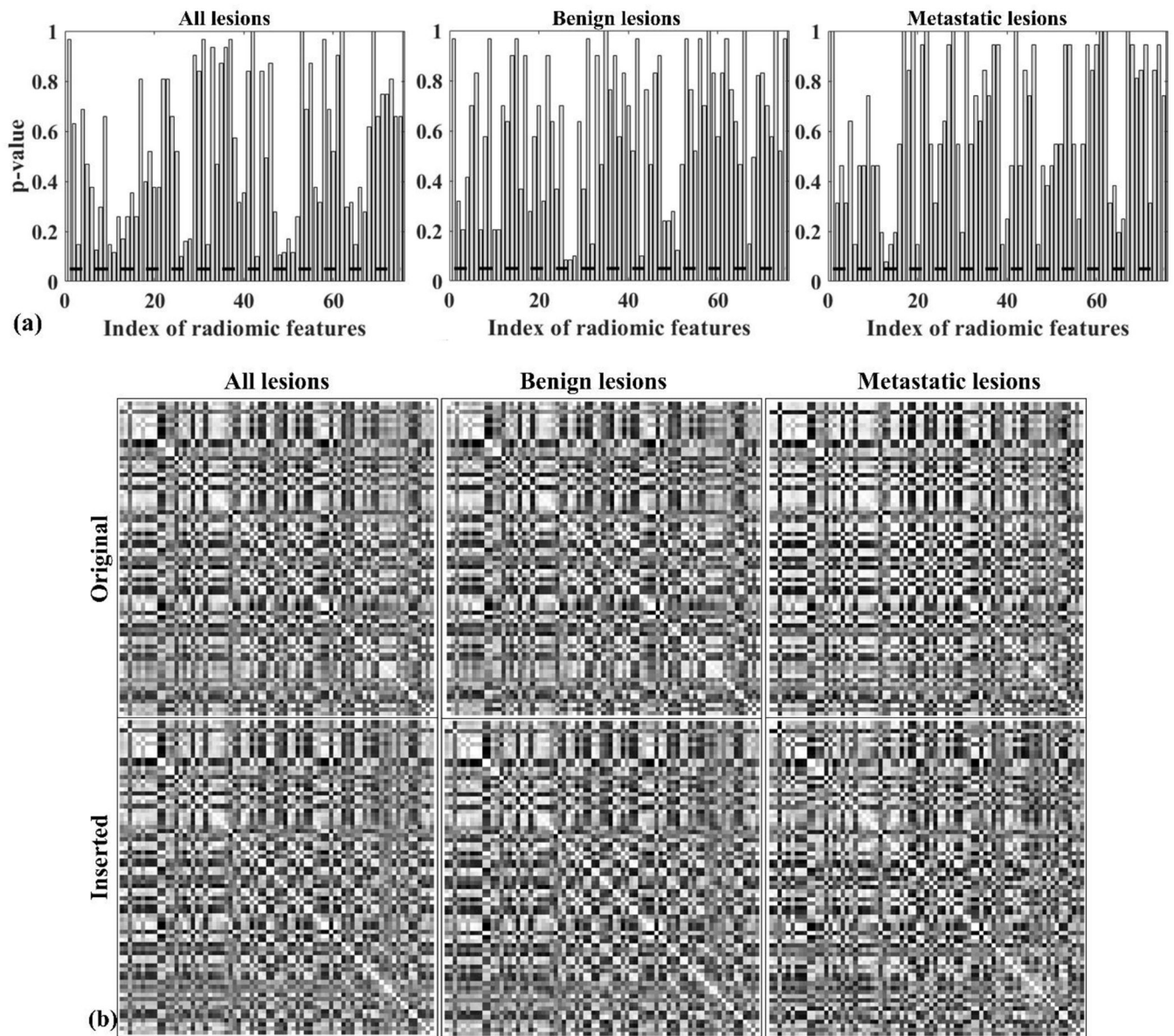


Figure 5.

(a) Wilcoxon signed rank test of radiomic features ($n=75$) at original and re-inserted locations. The dashed line indicates the level at $p=0.05$. (b) Radiomic feature correlation matrices from original and re-inserted lesions. Analyses were conducted over all lesions as well as separately for benign and metastatic sub-groups.

Table 1
Major parameters in patient scan and image reconstruction

Scanner	NAEOTOM Alpha (Siemens)
Tube voltage (kV)	120
CTDIvol (mGy)	9.8 ± 4.8
CARE keV IQ level	180
Helical pitch	0.6
Reconstruction algorithm	Quantum Iterative Reconstruction (QIR)
QIR strength	3
Reconstruction kernel	Br44
Field-of-view (mm)	200
Reconstructed image types *	Low-energy threshold, high-energy threshold, and 65 keV VMI
Image matrix size	512×512
Slice thickness / increment (mm)	3 / 1

* Image reconstructions used x-ray signals above low- and high-energy thresholds, respectively, which were analogous to conventional dual-energy CT. Virtual monoenergetic images (VMIs) at 65keV images were also reconstructed.

Table 2

Summary of patient cases and lesions

	Age	BMI [†]	Lesion size (mm)	Lesion contrast (HU)
Mean ± STD *	69.1±12.8	25.8±4.7	16.0±7.3	45.0±24.9
Range	[50, 85]	[20, 34.9]	[7.4, 39.5]	[14.4, 98.6]

^{*} STD – standard deviation[†] BMI – body mass index

Table 3

Mean CT numbers of different rods at original and re-inserted locations *

	Adipose	Blood 40 †	Blood 100 †	Iodine 2mgI/cc	Iodine 5mgI/cc	Calcium 100mgCa/cc	Calcium 300mgCa/cc
Original (TL)	-70.3	46.3	110.6	59.8	134.9	346.9	896.2
Inserted (TL)	-67.6	46.4	107.8	59.2	133.8	344.1	894.5
HU difference (TL)	-2.7	-0.1	2.8	0.6	1.1	2.8	1.7
% difference (TL) ^o	3.8%	-0.2%	2.5%	1.0%	0.8%	0.8%	0.2%
Original (TH)	-63.4	41.3	107.4	38.7	82.8	293.5	739.2
Inserted (TH)	-60	42.0	103.0	37.5	83.9	295.5	740.9
HU difference (TH)	-3.4	-0.7	4.4	1.2	-1.1	-2.0	-1.7
% difference (TH) ^o	5.4%	-1.7%	4.1%	3.1%	-1.3%	-0.7%	-0.2%

* TL & TH – Image reconstruction using signals from above lower and higher energy thresholds, respectively; Wilcoxon signed rank test showed no statistically significant difference ($p>0.05$); For each rod, CT number was averaged across >100 axial images to minimize noise perturbation, using one circular region-of-interest (with >20mm diameter) per image.

† Blood 40 and Blood 100: official label names from the phantom manufacturer. These two inserts mimicked blood samples with different iron densities.

^o % difference: the ratio between HU difference (original minus inserted) and the original HU.

Table 4

SSIM and mean CT number difference between original and re-inserted lesions

	All lesions		Benign lesions		Metastatic lesions	
	Mean $\pm$ STD *	Range	Mean $\pm$ STD *	Range	Mean $\pm$ STD *	Range
SSIM	0.95 $\pm$ 0.02	[0.93, 0.98]	0.95 $\pm$ 0.02	[0.93, 0.98]	0.96 $\pm$ 0.01	[0.94, 0.98]
HU difference †	-1.8 $\pm$ 6.4	[-12.8, 11.1]	-2.0 $\pm$ 6.6	[-12.8, 5.97]	-1.6 $\pm$ 5.6	[-8.0, 11.1]
% difference	-1.1% $\pm$ 6.4%	[-11.7%, 11.0%]	-0.9% $\pm$ 7.4%	[-11.7%, 11.0%]	-1.3% $\pm$ 5.2%	[-9.0%, 8.7%]

* STD – standard deviation

† Wilcoxon signed rank test showed no statistically significant difference (p>0.05)